Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/03/2013
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	Flight Support and Data Analysis for the CVB Microgravity Flight Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	07/15/2009	**End Date:**	09/30/2013
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:	ISS		
Flight Assignment:	ISS Increments 23/24 , 25/26 NOTE: Extended to 9/30/13 per PI and NSSC information (Ed., 8/5/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wayner, Peter (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX09AL98G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Constrained Vapor Bubble Experiment (CVB) is studying the fundamental physics and interfacial phenomena underlying the operation and performance of a miniature, wickless heat pipe in microgravity. The goal of the research is to understand the fluid mechanics and heat transfer processes occurring in these devices as we evaporate and condense the working fluid, pentane. The practical objective of the work is to improve the design of future space-based thermal control systems. In addition, the research will help improve the design of terrestrial heat pipes for use in laptops, servers, high-power LED luminaires, and even for the treatment of epileptic seizures. The CVB is composed of a glass heat pipe fabricated from a conventional spectrophotometer cuvette. An optical technique is used to measure the shape of the vapor-liquid interface over the entire surface of the cuvette. Thermocouples are used to measure the temperature gradient along the main axis of the heat pipe as well as several off-axis points. By measuring the shape of the vapor-liquid interface we calculate the capillary forces driving fluid flow from the cooled to the heated end of the system and by measuring the temperature profile we also calculate the heat transfer throughout the system. Thus we obtain a complete description of the operation of the device and can literally "see" why it operates as it does on Earth and in microgravity. NOTE: this is a continuation of NNC05GA27G and NAG3-2383 (PI=Wayner).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this last experiment will help to improve the design of all devices that rely on changing the phase of a fluid such as boilers, distillation columns, and refrigeration systems.
Task Progress:	During the past year of the research, we have been vigorously working on analyzing the data from the flight experiments performed during 2010, writing papers, and presenting the information at as many conferences as we could reasonably attend. We have also been preparing for CVB-2, a re-flight of the 30 mm CVB module that will be filled with a binary fluid mixture. The highlights are listed below: 1. We have automated, as much as possible, the analysis of surveillance camera data, fringe image data, and pressure and temperature data for the CVB modules. We can now perform real-time data analysis as the experiments are running. 2. We have established and finalized, with GRC, all experimental procedures, the experiment schedules, how we organize ourselves and communicate during the operation, and how we store and catalog all the data to insure maximum use of on-orbit operations time. 3. We extended our mathematical analyses to include an improved, global (1-D) fluid flow and heat transfer model for describing overall CVB performance. 4. We have developed a new finite element model to predict the film thickness profile and heat flux throughout a meniscus. This model has been extended to include temperature and concentration dependent fluid properties, enhanced expressions for surface curvature, the ability to represent perfectly and partially wetting fluids, and the ability to incorporate a second fluid component so that we can investigate the effect of temperature and concentration driven Marangoni flows. 5. We have developed a new model for meniscus oscillations that predicts when the meniscus should be stable and are developing new equipment that will allow us to capture oscillations with much better fidelity than we can now achieve. 6. During 2010, we performed flight experiments using all four of the CVB-1 modules. The 30 mm module was run in March 2010, the dry module in April, the 40 mm module in July, and the 20 mm module twice, once in June and again in late October. All were unqualified successes. A more detailed description of events can be found in the two LIS reports we submitted for the project. 7. We recorded some unexpected behavior with the 20 mm module on its first run, single bubble nucleation events, that we further investigated and characterized during the second set of 20 mm runs. We thank all those involved in ISS scheduling for giving us all the extra run time. 8. We have analyzed all the temperature, pressure, and 10x objective data from the experiments and generated complete mappings of the fluid profiles within each of the cells. 9. The dry, 30 mm, and 40 mm modules were returned to Earth. The liquids were drained from the modules and analyzed for contamination. No significant second fluid contamination was observed. Data analysis suggests that the glass nanoparticles introduced into the cell during the fabrication process affects phase change processes at the heater end. This needs further evaluation since there presence may be beneficial. The 30 mm and 40 mm modules were refurbished and refilled for CVB-2 experiments. 10. We determined that the heat pipes operate more efficiently in a microgravity environment. The internal fluid recirculation is greater, the internal heat transfer coefficients are about an order of magnitude greater in microgravity, and the temperatures at the heater interface are all lower in microgravity. 11. Due to the lack of natural convection in microgravity, the heat pipes run hotter and at higher internal pressures than they do in a 1-g environment. 12. We graduated one PhD student, Arya Chatterjee and he is currently employed at Intel. We now have a second graduate student, Akshay Kundan who is scheduled to graduate in another 3 years. 13. We prepared and submitted a proposal for CVB-2, established the working fluids and phase ratios required for this new experiment, prepared a new SRD for this experiment, and completed the SCR and RDR evaluations. The experiment is aboard the ISS awaiting running in Fall 2013. 14. We have been analyzing 50x data from CVB-1 though in general, that data is not of as high a quality as we get on the ground so we will have limited results. We are redesigning the focusing algorithms and image quality criteria needed to get the best data. This may reduce the overall scope of the 50x experimental matrix for CVB-2 but hopefully will ensure that most of the 50x data will be usable.

	15. We have been planning for CVB-3, looking at transitioning from a straight to a loop heat pipe configuration that will yield more practical results yet still be able to provide the fundamental science needed to investigate how those heat pipes perform, what geometries are required for best operations and what types of liquids and liquid mixtures should be charged into the heat pipe. 16. We have quantified some interesting results from CVB-1 regarding the magnitude and extent of internal radiation heat transfer. These results appear to have implications for what might be required to induce dryout of the module in microgravity. They will be the subject of a future paper later in 2013 and are the subject of a presentation in November. Submitted Abstract for 2013 ASGSR Annual Meeting, Orlando, FL, 3-8 Nov., 2013: CVB: The Constrained Vapor Bubble Capillary 40 mm Fin Experiment on the ISS. Peter C. Wayner, Jr., Akshay Kundan and Joel L. Plawsky. 17. Overall, we have 20 publications and have given 19 presentations. We are preparing an additional paper as mentioned above and have at least one or more presentations scheduled for 2013. 18. Conducted tests of current design of LMM/CVB-2 at NASA GLENN and Zin-Tech during the period 1/10/13 – 1/11/13. We did a preliminary analysis of this data. Sent LTR 1) Joel L. Plawsky and Peter C. Wayner, Jr., “CVB-2 Flight Status” (1/18/2013 letter/e-mail Evaluation of Flight Readiness of CVB-2 Based on 1g Data). We recommended changes and preparation of flight module for additional 1g testing scheduled to start January 29. 19. The new graduate student has re-started experimental work in our laboratory at RPI. This research will be used to test the effect of the composition of a second component on Marangoni Flow. We also plan to try some initial designs of a loop heat pipe configuration.
Bibliography Type:	Description: (Last Updated: 10/14/2025)
Articles in Peer-reviewed Journals	Chatterjee A, Plawsky JL, Wayner PC Jr, Chao DF, Sicker RJ, Lorik T, Chestney L, Margie R, Eustace J, Zoldak J. "Constrained vapor bubble heat pipe experiment aboard the International Space Station." Journal of Thermophysics and Heat Transfer. 2013 Apr;27(2):309-19. http://dx.doi.org/10.2514/1.T3792 , Apr-2013
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Articles in Peer-reviewed Journals	Kundan A, Plawsky JL, Wayner Jr PC, Chao DF, Sicker RJ, Motil BJ, Lorik T, Chestney L, Eustace J, Zoldak J. "Thermocapillary phenomena and performance limitations of a wickless heat pipe in microgravity." Physical Review Letters. 2015 Apr 10;114(14):146105. http://dx.doi.org/10.1103/PhysRevLett.114.146105 , Apr-2015
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